

$$C_{\phi B}^{(21)} \rightarrow$$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,0} [m_{\tilde{d}}^r, m_{\tilde{q}}^p] - \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,0} [m_{\tilde{d}}^r, m_{\tilde{q}}^p] + \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \right. \\ & \quad \text{LF}_{3,2,-1} [m_{\tilde{d}}^r, m_{\tilde{q}}^p] + \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{4,1,-1} [m_{\tilde{d}}^r, m_{\tilde{q}}^p] - \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,2,0} [m_{\tilde{e}}^r, m_{\tilde{l}}^p] - \\ & \quad \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,0} [m_{\tilde{e}}^r, m_{\tilde{l}}^p] + \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,2,-1} [m_{\tilde{e}}^r, m_{\tilde{l}}^p] + \\ & \quad \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{4,1,-1} [m_{\tilde{e}}^r, m_{\tilde{l}}^p] + \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,2,-1} [m_{\tilde{l}}^p, m_{\tilde{e}}^r] - \\ & \quad \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{4,1,-1} [m_{\tilde{l}}^p, m_{\tilde{e}}^r] + \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{5,1,-2} [m_{\tilde{l}}^p, m_{\tilde{e}}^r] + \\ & \quad \frac{1}{3} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,0} [m_{\tilde{q}}^p, m_{\tilde{d}}^r] + \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,2,-1} [m_{\tilde{q}}^p, m_{\tilde{d}}^r] - \\ & \quad \frac{13}{12} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{4,1,-1} [m_{\tilde{q}}^p, m_{\tilde{d}}^r] + \frac{3}{4} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{5,1,-2} [m_{\tilde{q}}^p, m_{\tilde{d}}^r] - \\ & \quad \frac{1}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,0} [m_{\tilde{q}}^p, m_{\tilde{u}}^r] - \frac{1}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,0} [m_{\tilde{q}}^p, m_{\tilde{u}}^r] + \\ & \quad \frac{1}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,2,-1} [m_{\tilde{q}}^p, m_{\tilde{u}}^r] + \frac{1}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{4,1,-1} [m_{\tilde{q}}^p, m_{\tilde{u}}^r] - \\ & \quad \frac{7}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,0} [m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \frac{1}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,2,-1} [m_{\tilde{u}}^r, m_{\tilde{q}}^p] - \\ & \quad \frac{11}{24} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{4,1,-1} [m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \frac{3}{4} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{5,1,-2} [m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \\ & \quad \frac{1}{8} m_1 \tilde{\mu} g_1^4 \text{LF}_{3,1,0} [\tilde{\mu}, m_1] - \frac{1}{4} m_1 \tilde{\mu} g_1^4 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] + \frac{1}{4} m_1 \tilde{\mu} g_1^4 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] + \\ & \quad \left. \frac{3}{8} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{3,1,0} [\tilde{\mu}, m_2] - \frac{3}{4} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] + \frac{3}{4} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] \right) \end{aligned}$$